

N = 136 olhos

Brazilian Cohort

Freedom Biotech

Progressive Corneal Remodeling até 9 Meses

Evaluation of changes induced by the Ferrara® HM ring: evidence of continuous flattening and active stromal remodeling well beyond 3 months postoperatively.

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Clinical Data — Temporal Progression

5.05 D

ΔKMAX (9M)

4.94 D

ΔKM (9M)

+42.8 μm

ΔPACHYMETRY

+0.79

ΔQ (9M)

Flattening Progression (ΔKmax)

Pré→3m

0.35

3m→9m

+4.70 D

Total 9m

5.05 D

⚡ 93% of total Kmax reduction occurs AFTER the 3rd month — progressive stromal remodeling

Summary Freedom (all HM)

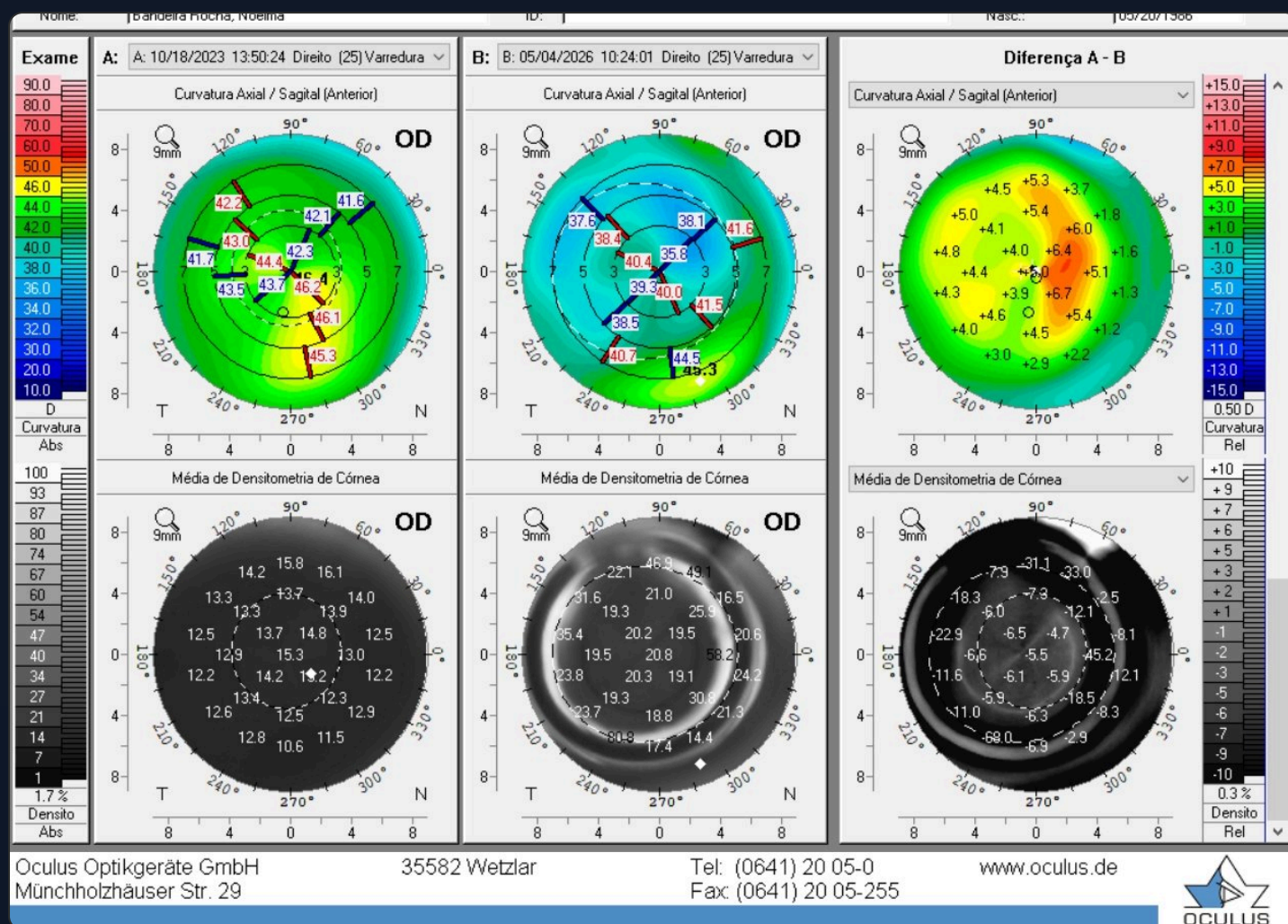
PARAMETER	Δ PRÉ→3M	Δ 3M→9M	Δ TOTAL 9M
K1	1.47	2.85	4.32 D
K2	2.09	3.27	5.36 D
KM	1.92	3.01	4.94 D
Kmax	0.35	4.70	5.05 D
Q	0.42	0.37	0.79
Pachymetry	+9.58	+33.19	+42.77 μm

By Model (3m PO)

MODEL	N	ΔK1	ΔKMAX	ΔASTIG
HM 300	23	2.92	6.92 ★	—
HM 350	6	3.75	4.58	1.13
HM 210	3	2.04	4.81	1.40
HM 250	8	1.49	2.69	1.45
HM 400	5	0.60	1.91	1.60

Clinical Cases

Case 1 — Keratoconus (OD) | Pentacam Compare



Anterior axial curvature pre vs post. Flattening of +3.9 to +6.7 D in the cone zone.

Scheimpflug — HM Ring In Situ



Transparency preserved. Adequate centration. No deposits/neovascularization.

| Advantages of Ferrara® HM



Fusiform Profile

Exclusive fusiform. 320° arc, Ø 5.7 mm.



Progressive

93% effect after 3m. Up to 9+ months.



Concentric

Double-ring for advanced KC.



+ Refractive

PRK/LASIK in selected cases.



HM 200-450

Mild to advanced KC + myopia.

BR

100% Brazilian

Pioneer 1986. Freedom.

| Conclusions

Progressive remodeling: ΔK_{max} 5.05 D at 9m, 93% after 3m.

Tissue remodeling: +42.77 μm = active stroma, not just mechanical.

HM 300 highlight: ΔK_{max} 6.92 D (n=23) — above literature.

Versatilidade: Concentrics + sinergia refrativa ampliam indicações.

References:

1. Ferrara P, et al. J Cataract Refract Surg. 2019.
2. Piñero DP, Alcón JL. Eye Vis. 2022.
3. Vega-Estrada A, Alió JL. J Refract Surg. 2016.

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FERRARA OPHTHALMICS

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Data for scientific and educational purposes. Individual results may vary.